WEEK 09 | OBJECT DETECTION I

YCB-V full-frame object detection - Faster R-CNN baseline

Intern	Mehrana Rezaei	Group	Group 1 - Basic ML Knowledge
Main Quest	Convolutional Neural Networks	Side Quest	YCB-Video Object Detection I

1. Objective

The practical goal was to move from the cropped-object classifiers developed in Weeks 03-08 to **full-frame object detection** on YCB-Video. Instead of receiving a ground-truth crop, the model predicts each selected object's class and bounding-box coordinates from the complete RGB scene. This week establishes an initial detection pipeline and evaluates localization using Intersection over Union (IoU).

2. Main Quest concepts and their application

CNN concept	How it appears in the executed pipeline
Pretrained feature extraction	COCO-pretrained MobileNetV3 backbone and FPN extract multi-scale image features.
Region proposal and classification	Faster R-CNN proposes object regions; the ROI head predicts foreground classes and box offsets.
Transfer learning	The pretrained detector is fine-tuned after replacing its classification/regression predictor with a six-class head.
Localization and evaluation	Class-aware one-to-one IoU matching, AP50, precision, recall and F1 measure detection quality.

3. Dataset and scene-level split

The same YCB-V BOP'19 subset and five object categories were retained: tomato_soup_can (4), sugar_box (3), master_chef_can (1), tuna_fish_can (6), and bleach_cleanser (12). Frames were sampled deterministically, at most 50 per scene, rather than sampling object crops.

Split	Scene IDs	Frames	Selected boxes
Train	48, 49, 50, 53, 54, 56, 57, 58	400	584
Validation	52, 55	100	250
Repeated test	51, 59	100	300

Data integrity: split scenes are disjoint and all five categories appear in every split. The executed dataset summary reports 16 training frames without selected target objects. Other YCB categories are visible but not foreground-labeled (1,150 instances in training, 200 in validation, 250 in test); 16 additional selected training instances were removed by visibility/box-size filtering. This makes the five-class annotations incomplete at the full-scene level.

4. Method and implementation

Stage	Observed implementation
Source / annotations	Hugging Face BOP archives; parse scene_gt.json and scene_gt_info.json with RGB frames.
Detection target	BOP bbox_obj (fallback bbox_visib), clipped xyxy boxes; visibility at least 0.35 and sides at least 24 px.
Model	Faster R-CNN with MobileNetV3-Large FPN; pretrained COCO weights; new six-output predictor (five classes plus background).
Training	T4 GPU, PyTorch 2.11.0 / Torchvision 0.26.0, batch size 2, AdamW LR 0.0001, weight decay 0.0001, AMP.
Image / model setup	Full RGB inputs around 640 x 480; detector internal resizing min 480 / max 640; no GT crop at inference.
Selection	Save the checkpoint with the highest validation mAP@0.50. Test scenes are evaluated after selecting the checkpoint.

4.1 Recorded training progress

The notebook was configured for eight epochs. Its saved outputs contain **six completed epochs**; epoch 7 started but did not finish in the saved log, and epoch 8 is not evidenced. The best saved checkpoint was selected at **epoch 4**, with validation mAP50 = **0.6970**. The table below transcribes the six completed epoch lines printed by the notebook.

Epoch	Train loss	Val mAP50	Val precision	Val recall
1	1.0770	0.6929	0.703	0.740
2	0.7110	0.6863	0.572	0.764
3	0.5962	0.6954	0.632	0.796
4	0.5555	0.6970	0.589	0.804
5	0.5441	0.6629	0.522	0.672
6	0.2648	0.6794	0.589	0.768

4.2 Full-frame ground-truth audit



Figure 1. Original full-frame bounding-box examples from the executed notebook, one from each split.

5. Results

Validation and repeated-test metrics are taken directly from the final evaluation cell of the executed notebook. The confidence threshold for operating-point precision/recall is 0.50; a prediction counts as a true positive only if it is the highest-ranked unmatched same-class prediction with IoU at least 0.50. AP50 uses the confidence ranking above the model's 0.05 floor.

Validation mAP@0.50	Repeated-test mAP@0.50	Best epoch
0.6970	0.8671	4

Measure	Validation (100 frames)	Repeated test (100 frames)
mAP@0.50	0.6970	0.8671
Precision @ score 0.50	0.5894	0.8498
Recall @ score 0.50	0.8040	0.7733
F1 @ score 0.50	0.6802	0.8098
Matched-object mean IoU	0.8291	0.8481
TP / FP / FN	201 / 140 / 49	232 / 41 / 68

Interpretation note: mean matched IoU is computed only across true-positive matches, not all objects. Test mAP50 is higher than validation mAP50, but these are different scenes and the test scenes were examined in earlier weeks. The difference does not prove a model improvement or an unbiased generalization gain.

5.1 Class-specific performance

The following two tables reproduce the class-level evaluation values displayed in the executed notebook.

Validation (100 frames)

Class	TP	FP	FN	Precision	Recall	AP50
tomato_soup_can	50	53	0	0.4854	1.00	0.9874
sugar_box	44	2	6	0.9565	0.88	0.9701
master_chef_can	50	50	0	0.5000	1.00	0.5000
tuna_fish_can	50	0	0	1.0000	1.00	1.0000
bleach_cleanser	7	35	43	0.1667	0.14	0.0274

Repeated held-out test (100 frames)

Class	TP	FP	FN	Precision	Recall	AP50
tomato_soup_can	50	7	50	0.8772	0.50	0.6592
sugar_box	50	5	0	0.9091	1.00	1.0000
master_chef_can	50	0	0	1.0000	1.00	1.0000
tuna_fish_can	50	15	0	0.7692	1.00	0.9992
bleach_cleanser	32	14	18	0.6957	0.64	0.6770

5.2 Representative validation detections

The following full-frame overlays are embedded from the executed notebook. Ground-truth boxes are on the left; the model's predictions at score 0.50 are on the right. They illustrate that the detector can localize selected classes, but also predicts some selected categories on additional visible objects that were not among the five positive labels.

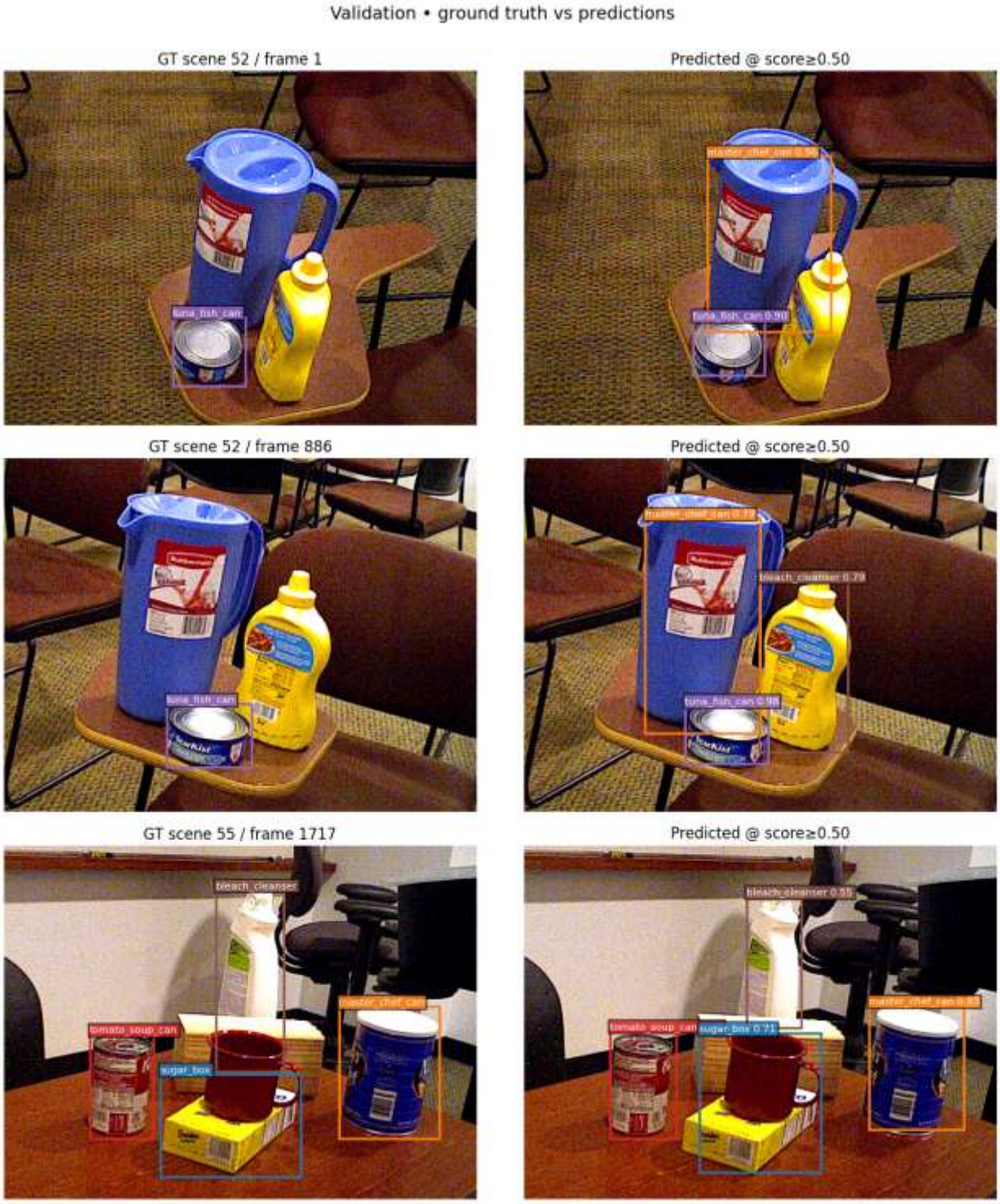


Figure 2. Actual executed validation examples from scenes 52 and 55 (GT versus predicted boxes).

5.3 Representative repeated-test detections

The same fixed detector and confidence threshold are shown on scenes 51 and 59. Some selected objects are accurately localized, while specific objects are missed: in the second example, the visible tomato_soup_can has a ground-truth box but no qualifying prediction. These are qualitative examples, not a separate selection procedure.



Figure 3. Actual repeated-test examples from the executed notebook. This is not an official BOP benchmark evaluation.

6. Failure analysis

Class-wise evidence. On validation, bleach_cleanser recall was only 0.14 (7 true positives and 43 false negatives), and its AP50 was 0.0274. Tomato_soup_can generated 53 false positives and master_chef_can generated 50, despite perfect recall for both. On the repeated test split, tomato_soup_can recall fell to 0.50 (50 misses among 100 annotated objects); the strongest test recall values were obtained by sugar_box, master_chef_can and tuna_fish_can (all 1.00).

Visual evidence. The hardest validation frames in the saved error table are concentrated in scene 55 (including frames 62 and 526, each with two false positives and two false negatives). The examples show missing bleach and sugar boxes along with extra tomato_soup_can predictions around a red cup and nearby objects. This pattern is consistent with background/context confusions and partial labels, but neither cause has been experimentally isolated.

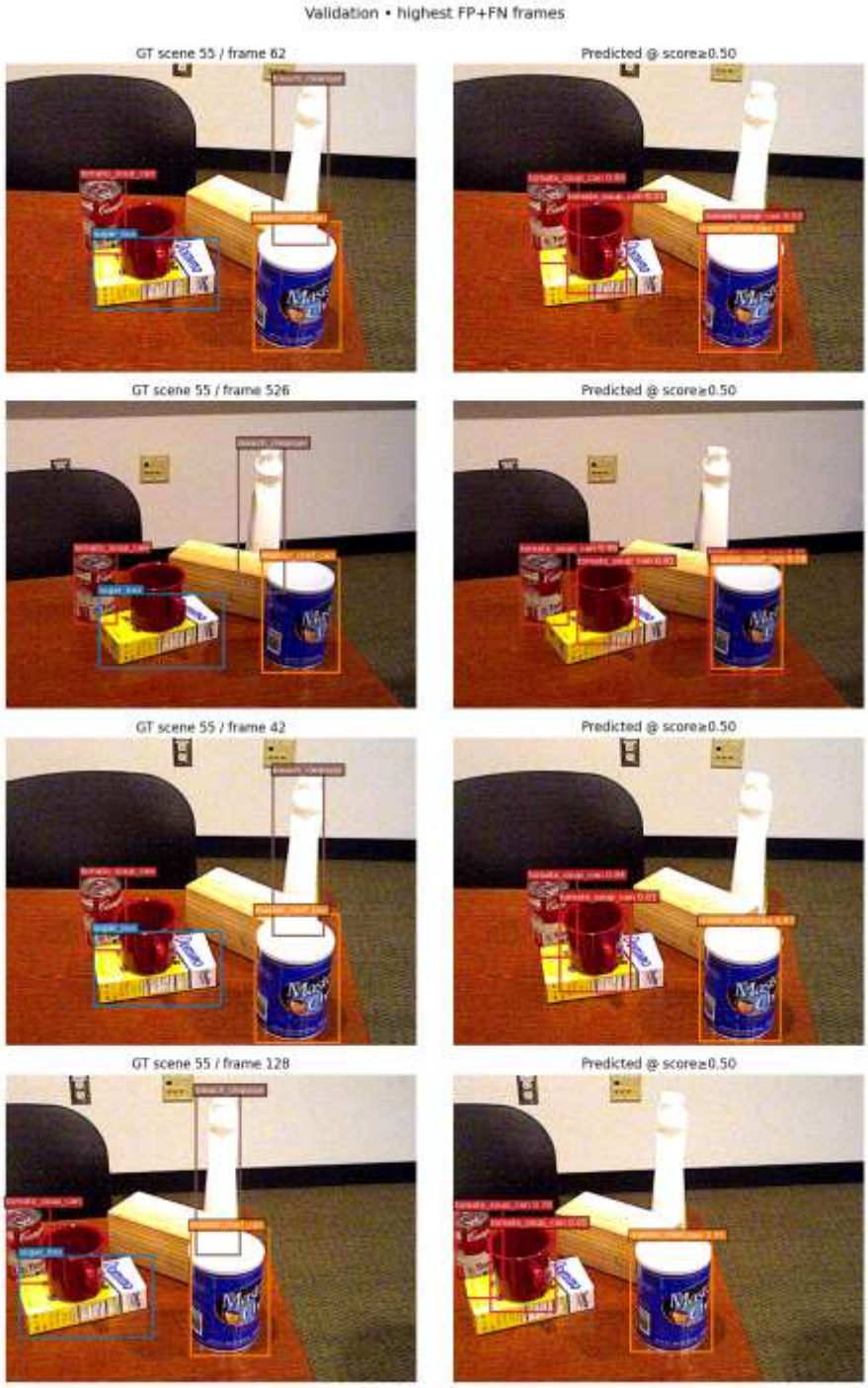


Figure 4. Actual highest-error validation examples (ground truth at left; predictions at right).

7. Conclusion and Week 10 plan

Week 09 established a working full-frame five-class detection pipeline with frame-level BOP annotations, COCO-pretrained Faster R-CNN fine-tuning, validation-only checkpoint selection, class-aware IoU matching, and saved qualitative predictions. The selected epoch-4 model achieved validation mAP@0.50 of **0.6970** and repeated-test mAP@0.50 of **0.8671**. Unlike Weeks 03-08, these numbers evaluate predicted bounding boxes and categories together and must not be compared directly with crop-classification accuracy.

The detector is not yet consistently reliable across classes. The validation bleach_cleanser result is the clearest weakness; test tomato_soup_can recall and excess validation detections are also priorities. Scene-specific differences, unannotated non-target objects, limited sampling, and a single training seed limit the strength of generalization claims.

Recommended Week 10 controlled experiments

Priority	Experiment and evidence to collect
1. Confidence thresholds	Sweep thresholds on validation only; report class-wise precision/recall curves and operating-point tradeoffs.
2. Duplicate boxes / NMS	Inspect overlapping predictions and vary NMS settings one factor at a time; track duplicate-related FP counts.
3. Partial-label audit	Manually verify apparent FPs that overlap non-selected YCB categories; distinguish true detector mistakes from annotation scope.
4. Stronger evaluation	Keep fixed scenes for historical comparison but reserve fresh scenes or a new split for later final assessment.